

Initial Project Planning Template

Date	13 July 2025
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection	USN-1	As a developer, I can collect and review the application_record.csv and credit_record.csv datasets to understand the available applicant and credit-history fields.	2	High	Reddy Venkata Surya Chakra Mahalakshmi	03-07-2026	06-07-2026
Sprint-1		USN-2	As a developer, I can perform exploratory data analysis (EDA) to understand data distributions, class balance, and missing values.	3	High	Reddy Venkata Surya Chakra Mahalakshmi	03-07-2026	06-07-2026
Sprint-1	Data Preprocessing	USN-3	As a developer, I can clean the data (remove duplicates, handle missing OCCUPATION_TYPE values) and engineer new features such as Family_Dependency.	3	High	Reddy Venkata Surya Chakra Mahalakshmi	03-07-2026	06-07-2026
Sprint-1		USN-4	As a developer, I can encode categorical features (gender, car/realty ownership, income type, education, family status, housing type) using label encoding.	2	Medium	Reddy Venkata Surya Chakra Mahalakshmi	03-07-2026	06-07-2026
Sprint-2	Model Building	USN-5	As a developer, I can merge the application and credit-history datasets and convert the STATUS field into a binary approval label.	2	High	Reddy Venkata Surya Chakra Mahalakshmi	07-07-2026	10-07-2026

Sprint-2		USN-6	As a developer, I can train and compare multiple classification models (Logistic Regression, Decision Tree, Random Forest) on the processed data.	5	High	Reddy Venkata Surya Chakra Mahalakshmi	07-07-2026	10-07-2026
Sprint-2		USN-7	As a developer, I can evaluate each model using confusion matrix, classification report, and F1-score, and select the best-performing model.	3	High	Reddy Venkata Surya Chakra Mahalakshmi	07-07-2026	10-07-2026
Sprint-3	Application Development	USN-8	As a developer, I can serialize the selected model (Random Forest) using joblib for use in the web application.	1	High	Reddy Venkata Surya Chakra Mahalakshmi	11-07-2026	13-07-2026
Sprint-3		USN-9	As a developer, I can build a Flask web application with a form to collect applicant details for prediction.	3	High	Reddy Venkata Surya Chakra Mahalakshmi	11-07-2026	13-07-2026
Sprint-3		USN-10	As a user, I can submit my applicant details through the web form and view an Approved/Rejected prediction result.	2	High	Reddy Venkata Surya Chakra Mahalakshmi	11-07-2026	13-07-2026
Sprint-4	Testing & Deployment	USN-11	As a developer, I can test the /predict endpoint with sample inputs to verify correct predictions and error handling.	2	Medium	Reddy Venkata Surya Chakra Mahalakshmi	11-07-2026	13-07-2026
Sprint-4		USN-12	As a developer, I can deploy the Flask application (via Gunicorn) so it is accessible as a live web app.	3	High	Reddy Venkata Surya Chakra Mahalakshmi	11-07-2026	3-07-2026